Six More Weeks of Overfitting: Stacked Rodent Networks for Seasonal Forecasting

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Abstract

For over a century, humanity has outsourced seasonal forecasting to a single stump-based binary classifier in Punxsutawney, Pennsylvania. NOAA's public analyses suggest this legacy system underperforms several peers and, in some years, exhibits strong "coin-flip energy" [3, 2]. We propose **Marmot-Stack**, a stacked ensemble that combines multiple groundhog prognosticators *using only their binary outputs as features*. We compile predictions from public groundhog archives and label outcomes using NOAA/NCEI March temperature anomalies; expanding-window walk-forward backtesting (with nested model selection) shows that a distributed rodent network reduces "Shadow Bias" and achieves measurable gains with minimal additional folklore.

1 Introduction

Groundhog Day is the only forecasting task in which (1) the model is furry, (2) the input is a shadow, and (3) the prediction is treated as both folklore and a press release. While the tradition is charming, the accuracy is suspect: NOAA's public-facing analyses routinely report that Punxsutawney Phil's recent accuracy is below 50% and below multiple alternative prognosticators [3, 2].

We ask a simple question: **can we do better by ensembling multiple groundhogs?**

In other domains this would be called stacked generalization [12, 13]. Here it is called inviting more rodents to the podium.

2 Data: Scraping the Stump Endpoints

2.1 Predictions (binary telemetry)

We scrape each groundhog's year-by-year prediction from `groundhog-day.com`, which provides a structured "All predictions" list for many named prognosticators (e.g., Punxsutawney Phil, Staten Island Chuck). Official Groundhog Prediction APIs remain undocumented and the free tier allows exactly one request per year. For each groundhog i and year t we record $S_{i,t} \in \{+1, -1\}$, where $+1$ denotes "More winter" and -1 denotes "Early spring" [1].

2.2 Endpoint proliferation over time

The number of participating groundhogs increases over time as more prognosticators enter the archives. Figure 1 shows the number of active groundhogs per year; we use this to define eras for within-era evaluation (see below).

3 Methodology: Regularizing Folklore

3.1 Task and labels

We cast Groundhog Day as a binary classification task. Each year t has a ground-truth label $y_t \in \{+1, -1\}$ derived from NOAA/NCEI March temperature anomalies: $y_t = -1$ ("early spring") if the March anomaly is positive, and $y_t = +1$ ("more winter") otherwise. We define anomalies using the NCEI *Climate at a Glance* national March series [11]. This choice mirrors NOAA's public-facing "grading" framing, which also evaluates prognosticators against U.S.

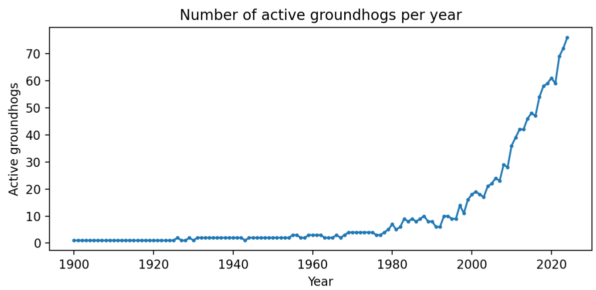

Figure 1: The Great Groundhog Proliferation: unique stump endpoints observed each year. The dataset is not i.i.d.; hype, local pride, and network effects keep spawning new endpoints.

March temperatures [2].

3.2 Model: Marmot-Stack

Each groundhog i outputs $S_{i,t} \in \{+1, -1\}$, with missing values mapped to 0 as abstentions. This treats absent endpoints as timeouts rather than votes, avoiding hallucinated shadows while letting the meta-model learn from whatever telemetry exists that year. We fit a regularized logistic meta-model (stacked generalization) over these outputs:

$$p(y_t = +1 \mid S_{\cdot,t}) = \sigma\left(b + \sum_i w_i S_{i,t}\right),$$

where σ is the logistic sigmoid. We predict $\hat{y}_t = \text{sign}(b + \sum_i w_i S_{i,t})$.[1] This model is intentionally boring; the animals already do the whimsical part.

Intuitively, some hogs are informative, some are noisy, and a select few are reliably contrarian ("anti-hogs"). In effect, we estimate how much predictive wood each woodchuck can chuck, subject to regularization.

3.3 Backtesting and model selection

We evaluate using expanding-window walk-forward backtesting: to predict year t, we train only on years $< t$. This avoids the common pitfalls of random k-fold CV under temporal dependence, where leakage can inflate performance estimates [6]. To avoid tuning on the future, we use *nested* walk-forward evaluation: within each training window, an inner time-respecting cross-validation chooses the regularization strength, and the outer loop evaluates on the held-out year [4, 5]. We report accuracy, balanced accuracy, and Matthews correlation coefficient (MCC) [8, 9]. We also compare against two baselines: always predicting early spring, and unweighted majority vote.

[1] If you prefer a threshold τ, set $\tau = -b$ and pretend this is a weighted vote. Same decision boundary, different vibes. Ties break toward "early spring."

4 Results: Less Phil, More Skill

Table 1 summarizes overall out-of-sample performance. Marmot-Stack improves accuracy over the always-early baseline by approximately 7 percentage points, and improves balanced accuracy and MCC, indicating that gains are not purely a class-imbalance artifact. The always-early baseline achieves 0.567 accuracy, reflecting class-prior drift toward "early spring" under our anomaly-based label. Unweighted majority vote collapses to 0.392 accuracy, consistent with a systematic bias toward predicting "more winter" across endpoints. Nested walk-forward improves to 0.642 accuracy (MCC 0.284), which is the most defensible estimate because regularization strength is chosen without peeking at the test year. Against the always-early baseline the evidence is suggestive (rather than decisive), which is exactly what one should expect from a 120-year dataset assembled from stump telemetry.

Because the class prior drifts over time, the practically relevant hypothesis test is not "better than 50%" but "better than always-early"; by that standard, the evidence is suggestive rather than a stump-splitting slam dunk. The gain is not enormous, but it is real enough to mildly inconvenience single-hog truthers.

4.1 Why majority vote fails

Majority vote underperforms dramatically because many groundhogs systematically over-predict "more winter" (Shadow Bias: a chronic tendency to predict "more winter" regardless of evidence). When the class prior shifts toward early spring, the crowd becomes confidently wrong. Marmot-Stack learns to down-weight—or, when necessary, invert—biased predictors.

4.2 Coverage drift and era behavior

Because the number of active groundhogs grows over time (Figure 1), we also report performance by activity-defined era (Table 2). The uplift over the always-early baseline remains modest but consistent across eras, suggesting the model's skill is not confined to a single historical regime.

5 Threats to Validity (Burrows & Confounders)

Our results are statistically non-embarrassing, but several burrow-level confounders remain. We enumerate them here so future researchers can either (a) fix them, or (b) cite this section and proceed anyway.

5.1 Label semantics and non-stationarity

Our primary label treats "early spring" as "March warmer than the NOAA/NCEI reference baseline" (positive March anomaly). This is convenient and NOAA-approved, but the baseline itself is not eternal: NOAA "climate normals" are typically 30-year averages and are periodically updated [10]. A fixed reference period can therefore induce class-prior drift over time. To avoid winning by merely memorizing the direction of climate change, we (i) report the always-early baseline, and (ii) include balanced metrics (balanced accuracy, MCC).

5.2 Coverage drift and survivorship bias

The dataset is not i.i.d.: the number of recorded prognosticators increases over time (Figure 1). Early years are effectively a two-hog regime, while later years resemble a distributed rodent network. This induces survivorship bias: long-lived, heavily recorded hogs have more opportunities to earn stable weights, while late arrivals have fewer training examples before they are asked to carry the hopes of spring.

We mitigate this by reporting performance by activity-defined eras and with rolling-window summaries computed on out-of-sample predictions only. If a hog was not yet in the archive, we do not blame it for missing data; we simply blame it for not being born in time for our train split.

5.3 Geography mismatch

We evaluate against a national March series, but many groundhogs inhabit local microclimates. A hog can be right about Alberta and wrong about the continental U.S.; we score this as "wrong," since our pipeline lacks a localization layer (and any form of maple-syrup regularization). Future work could label by state/region (e.g., NOAA climate divisions) while keeping the model feature-pure: only groundhog outputs are used as inputs; geography affects labels only.

5.4 Human-in-the-loop rodents

Groundhog Day is an ML system with a substantial human middleware layer. Predictions are announced at ceremonies, sometimes with theatrical constraints, and may incorporate non-rodent information (e.g., a weather forecast read by a handler, or a poem read by a handler who once saw a weather forecast). This can introduce hidden leakage (forecasting) or hidden noise (poetry). Since handlers publicly "translate" the prediction, our evaluation should be interpreted as measuring the end-to-end system (groundhog + ceremony) rather than rodent intelligence in isolation.

5.5 Model selection, p-values, and other root vegetables

We report binomial p-values versus a coin flip and versus the always-early baseline. These tests answer a narrow question: "is accuracy greater than p?" and do not correct for the many ways one can tune a marmot (label choice, era thresholds, window sizes, regularization, and which figure gets the funniest caption). We reduce selection bias via nested, time-respecting cross-validation [4, 5] and recommend permutation tests for end-to-end pipeline significance [7]. In short: the p-values are carrot-adjusted, but not fully radish-proof.

6 Conclusion

A single stump-based classifier is an obvious single point of failure (SPoF). By distributing inference across a stacked rodent network, we can measurably improve Groundhog Day forecasting while preserving the core cultural artifact: waking a sleepy animal up for no reason. In doing so, we upgrade the tradition from one charismatic endpoint to a modestly more reliable ensemble whose primary failure mode is now statistical rather than ceremonial.

To the best of our knowledge, this is the current

Table 1: Headline performance (walk-forward; $n = 120$ test years). "Always-early" predicts $\hat{y} = -1$ for every year.

Model	Accuracy	Bal. acc	MCC	p vs coinflip	p vs always-early
Always-early	0.567	—	—	—	—
Majority vote	0.392	0.357	-0.324	0.9933	—
Marmot-Stack (LogReg)	0.608	0.602	0.204	0.01104	0.2124
Marmot-Stack (nested CV)	0.642	0.643	0.284	0.001223	0.06126

Table 2: Metrics by era (nested walk-forward).

Era	n test	Accuracy	Bal. acc	MCC	Always-early	Δ vs early
high	41	0.805	0.668	0.408	0.756	+0.049
low	56	0.482	0.464	-0.075	0.429	+0.054
mid	23	0.609	0.642	0.321	0.565	+0.043

state of the art in rodent-based stacked ensemble networks under non-i.i.d. burrow logs: a narrow benchmark, but an important one.

Ethics Statement

No groundhogs were retrained, fine-tuned, or prompted beyond their usual shadow-related duties. We used only publicly available stump telemetry and purchased no shadow-broker data. No IRB was consulted; none returned our emails.

References

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